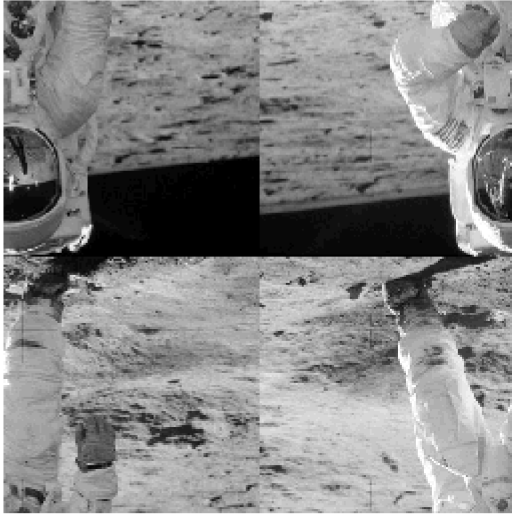


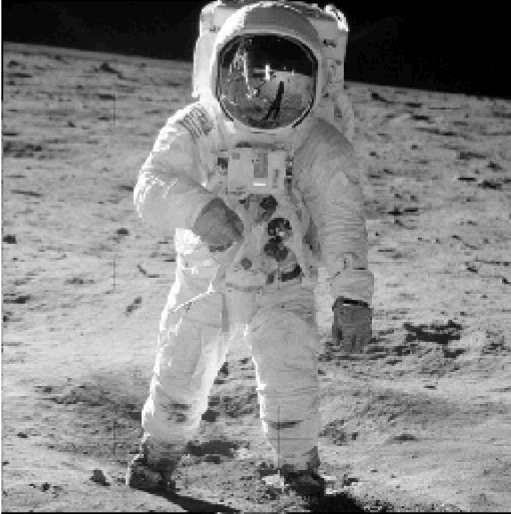
%% Task 1

```
compressed = imread("C:\Users\nrama\Downloads\Lab_06-1  
(1)\Lab_06\lab_06_image.tif");  
scaling_factor = 255/32;  
real = compressed * scaling_factor;  
imshow(real)
```



```
quad1 = flip(real(1 : 128, 1 : 128));  
quad2 = flip(real(1 : 128, 128 : 256));  
quad3 = flip(real(128 : 256, 1 : 128));  
quad4 = flip(real(128 : 256, 128 : 256));  
top = horzcat(quad2, quad1);  
bottom = horzcat(quad4, quad3);  
  
reconstruction = vertcat(top, bottom);  
figure  
imshow(reconstruction)  
title("Neil on the Moon! (recieved image scaled and rearranged)")
```

Neil on the Moon! (recieved image scaled and rearranged)



```
%% Task 2
elements = double(compressed(1:4, 1:4));
% Detrminant is a square matrix with a non-zero determinant meaning that it
% is nonsingular
det(elements);
solution_vector = [40; 30; 20; 10];
answers = elements \ solution_vector
```

```
answers = 4x1
    56.3158
   -70.0000
    76.3158
   -86.3158
```

```
% elements * answers = solution_vector
```

```
answers_sum = abs(sum(answers))
```

```
answers_sum = 23.6842
```

```
% The reception quality was excellent
```